1. Introduction to Pandas

Pandas is an open-source Python library for
data manipulation and analysis.

You can use Pandas to:

- See who bought the most
- Add a new column: "Did they tip?"
- Save your customer list to a file
- Find out the average number of lemonades bought

And you don't need to count with fingers –
Pandas does it with just one line of code!

Install and Import:

```
pip install pandas
```

```
import pandas as pd
```

```
pd.__version__
```

Key Data Structures:

Series = one list

DataFrame = full table

2. Pandas Series

Create Series:

```
sales = pd.Series([2, 3, 5])
```

Indexing and Slicing:

```
sales[1]



sales[0:2]
```

Attributes:

```
sales.index

sales.values

sales.dtype
```

Methods:

```
sales.head()

sales.tail()

sales.sort_values()

sales.count()

sales.isnull()
```

Operations:

```
sales + 1

sales * 2
```

Missing Values:

```
sales_with_missing = pd.Series([2, None, 5])

sales_with_missing.isnull()

sales_with_missing.fillna(0)
```

3. Pandas DataFrame

Create DataFrame:

```
data = pd.DataFrame({

    'Name': ['Anna', 'Ben'],

    'Lemonades': [2, 3]

})


s1 = pd.Series(data=[1,2,3,4,5])

s2 = pd.Series(data=[10,20,30,40,50])

si = pd.Series(data=[100,200,300,400,500],index=list('abcde'))


data = pd.DataFrame({'C1':s1, 'C2':s2})
```

Attributes:

```
data.shape

data.columns

data.dtypes
```

Accessing Data:

```
# Using labels

data.loc[1,'C1']


# Using index

data.iloc[1,0]
```

Methods:

```
data.head()

data.tail()

data.info()

data.describe()

data.sort_values()

data.count()

data.isnull()
```

Filtering:

```
data[data['Lemonades'] > 2]
```

Adding, Updating, Deleting:

```
data['Tips'] = [1, 2]


data.loc[data['Name'] == 'X', 'Y'] = 6


data.rename(columns={'Name':'NAMEE'})


del data['Tips']

data = data.drop(columns=['Lemonades'])
```

4. Indexing and Selection

Set Index:

```
data.set_index('Name', inplace=True)
```

Reset Index:

```
data.reset_index(inplace=True)
```

MultiIndex Example:

```
multi_data = pd.DataFrame({
    'Park': ['A', 'A', 'B', 'B'],
    'Name': ['Anna','Ben','Chris','Daisy'],
    'Lemonades': [2, 3, 4, 5]
}).set_index(['Park', 'Name'])
```

Selecting with Conditions:

```
multi_data[multi_data['Lemonades'] > 3]
```

Query Method:

```
multi_data.query('Lemonades > 3')
```

5. Working with Data

```
data.to_csv('lemonade_sales.csv')

data = pd.read_csv('lemonade_sales.csv')


dictt = {'x': [1, 2, 3], 'y': np.array([10, 20, 30]), 'z': 50}

pd.DataFrame(d)


listt = [[1, 2, 100], [2, 4, 100], [3, 8, 100]]

pd.DataFrame(listt, columns=['x', 'y', 'z'])


narr = np.array([[1, 2, 100], [2, 4, 100], [3, 8, 100]])

pd.DataFrame(narr, columns=['x', 'y', 'z'])
```

6. Data Cleaning and Preprocessing

Handling Missing Data:

```
data.dropna()

data.fillna(0)

data.isnull()
```

Remove Duplicates:

```
data.drop_duplicates()
```

Replace Values:

```
data.replace('lemondae', 'lemonade')
```

Type Conversion:

```
data['Lemonades'] = data['Lemonades'].astype(int)
```

String Operations:

```
data['Name'].str.upper()
```

7. Data Aggregation and Grouping

GroupBy Example:

```
grouped = data.groupby('Name')

grouped.sum()
```

Aggregations:

```
grouped['Lemonades'].mean()

grouped['Lemonades'].min()

grouped['Lemonades'].max()
```

Pivot Tables:

```
data.pivot_table(values='Lemonades', index='Name')
```

Crosstab:

```
pd.crosstab(data['Name'], data['Lemonades'])
```

8. Merging, Joining and Concatenating

Concatenation:

```
pd.concat([data1, data2])
```

Append:

```
data1.append(data2)
```

Merging:

```
pd.merge(data1, data2, on='Name')
```

Join on Index:

```
data1.join(data2.set_index('Name'), on='Name')
```

9. Working with Time Series

DateTime Index:

```
data['Date'] = pd.to_datetime(data['Date'])

data.set_index('Date', inplace=True)
```

Converting Strings to Dates:

```
pd.to_datetime('2025-04-29')
```

Resampling:

```
data.resample('W').sum()
```

Shifting/Lagging:

```
data['Previous Lemonades'] = data['Lemonades'].shift(1)
```

Rolling Calculations:

```
data['Rolling Average'] = data['Lemonades'].rolling(window=2).mean()
```

10. Data Visualization

Basic Plotting:

```
data['Lemonades'].plot()
```

Different ‘Kind’ of Plots:

```
data['Lemonades'].plot(kind='line')

data['Lemonades'].plot(kind='bar')

data['Lemonades'].plot(kind='barh')

data['Lemonades'].plot(kind='hist')

data['Lemonades'].plot(kind='box')

data['Lemonades'].plot(kind='scatter') #provide x&y axis data

data['Lemonades'].plot(kind='area')

data['Lemonades'].plot(kind='pie')
```

Customizations:

```
data['Lemonades'].plot(title='Lemonade Sales Over Time')
```

Pandas Notebook : Download here